

1. Definition of Dataset in Deep Learning:

dataset is a collection of **labeled** training example.

(labeling is a process creating a link between inputs (x) and desired outputs (y) [ground truth])

2. Garbage In Garbage Out (GIGO) Problem:

the dataset which neural networks learn from may suffer from human bias (or any other kind of bias).

3. Definition of Generalization in Deep Learning:

neural networks is not just provide correct predictions for data they have been trained on,

the more important goal is to provide correct predictions for data **previously unseen**.

4. Three Types of Datasets in Deep Learning:

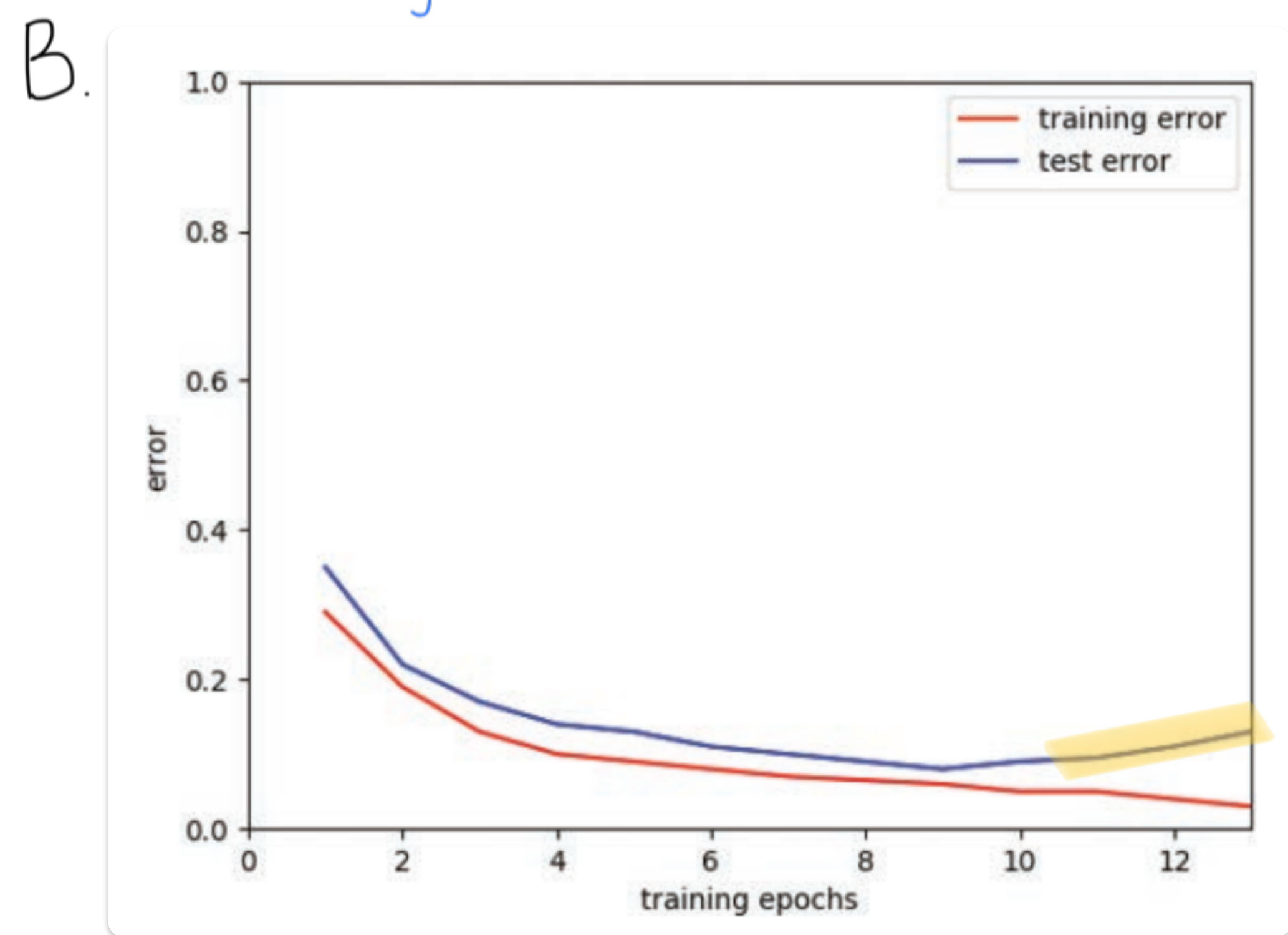
A. training dataset: it's used to tune parameters of neural networks and get **training error** during the training stage

B. validation dataset: it's used to tune hyperparameters of neural networks during the training stage

C. test dataset: it's used to evaluate how well neural networks are able to generalize to previously unseen data get **test error** during the training stage

5. Definition of Overfitting in Deep Learning:

A. the training error shows a downward trend until it finally flatten out while the test error, on the other hand, decreases in the beginning but then at some points starts increasing again, it shows a U-curve, this is a sign that the neural network is overfitting to training dataset.



6. Two of Approaches to Avoid Overfitting:

A. increase training dataset size to avoid overfitting.

B. employ regularization techniques (ex: early stopping) to avoid overfitting.

7. Concept of Early Stopping:

monitoring the test error and stopping when it starts to increase is difficult due to fluctuation of test error, which is not strictly moving in one direction, so we tend to stop and save weights of neural network at **fixed intervals**.

8. Parameter vs. Hyperparameter in Deep Learning:

A. parameter: it's a variable adjusted by perceptron learning algorithm automatically, (Ex: weight)

B. hyperparameter: it's a variable **not** adjusted by perceptron learning algorithm automatically, but adjusted by mankind manually, (Ex: learning rate (η), network architecture (network topology), types of activation function)

9. Information Leakage Problem:

if we tune hyperparameters on the basis of how neural networks performance on test dataset,

, then we introduced information leakage from test dataset, and the generalized evolution of neural networks will be too optimistic that has potential overfitting problem, but it doesn't reflect on the increase of test error due to the leakage.